

Week 3 Report

1. Context

Our problem statement (Week 1) is to predict which customers will accept a marketing campaign offer so the retailer can contact fewer people while still reaching most of the likely responders. It is a supervised binary classification problem. The data is the Customer Personality Analysis dataset (Kaggle): 2,240 customers and 29 columns covering demographics, two years of spend across six product categories, purchase channels, and acceptance of five earlier campaigns. The target, Response, is 1 if the customer accepted the sixth and most recent campaign. This report covers how we partitioned the data, the exploratory analysis we ran on the training set, what it tells us about the problem, and the pre-processing it calls for.

2. Dataset Partition Strategy

Strategy selected. We used a single 80/20 train/test split that is stratified on Response and group-aware, with a fixed random seed (42). The split was made before any EDA, and all analysis in this report uses the training set only. Model tuning in later weeks will use stratified, group-aware 5-fold cross-validation inside the training set; the test set will be opened once, for final evaluation.

Set	Rows	Share of data	Responders	Response rate
Full dataset	2,240	100%	334	14.9%
Training	1,792	80%	267	14.9%
Test (held out)	448	20%	67	15.0%

Table 1. Result of the stratified, group-aware split. No duplicate profile appears in both sets.

Why a held-out test set created before EDA. EDA is a form of fitting: the outlier caps, feature ideas and encoding choices we make here are all shaped by the data we look at. If the test set were

included, those choices would be tuned to it and its score would overstate real performance. Splitting first keeps the test set an honest proxy for new customers.

Why stratified. Only 14.9% of customers responded. A purely random 20% sample could easily land at 12% or 18% positives, which would distort both the class balance the model learns from and the precision and recall we report. Stratification fixes the rate at about 15% in both sets and guarantees 67 positives in the test set.

Why group-aware. Before splitting we checked for duplicate customers. Customer IDs are unique, but 394 rows share an identical profile on every feature with at least one other row, meaning the same customer was entered more than once under a different ID (201 extra copies). Nineteen of these duplicate profiles even carry conflicting Response labels. With a plain random split, copies of the same customer would end up in both train and test, and the model would be scored partly on customers it had already memorised. We assigned every distinct profile a group label and used scikit-learn's StratifiedGroupKFold, taking one of five folds as the test set, so each group falls entirely on one side. An assertion in the notebook confirms that no group is shared.

Alternatives we rejected. A three-way 70/15/15 train/validation/test split would leave about 50 positives in each of the validation and test sets, too few for stable estimates; cross-validation on the training set uses the data more efficiently. A time-based split does not fit this data: Response refers to a single campaign with no recorded date, and Dt_Customer is the enrollment date, not the date of the outcome, so there is no time order in the target to respect.

3. Exploratory Data Analysis

The Week 2 profiling used the full dataset (2,240 customers). All Week 3 analysis uses the training set only (1,792 customers, 267 responders).

3.1 Types of EDA performed and key results

Data quality checks. The data is mostly complete but contains hidden duplicates and a small number of implausible values.

- *Missing values.* Income is the only column with gaps: 24 in the full dataset, 16 in training. Too few to judge whether missingness relates to response.
- *Duplicates.* Customer IDs are all unique, but excluding ID and Response, 394 rows share an identical profile with another row (201 extra copies), and 19 profiles carry conflicting labels.

- In training, 318 rows belong to a duplicate group and 16 groups conflict.
 - Duplicated and unique rows respond at similar rates (15.7% vs. 14.7%), so the duplicates don't bias the target.
- *Constant columns.* Z_CostContact (always 3) and Z_Revenue (always 11) carry no information.
- *Potentially implausible values.*
 - Two training customers have birth years of 1899 and 1900. The third early birth year, 1893, is in the test set.
 - One income of \$666,666 belongs to a customer who spent only \$62. The next-highest income in the full dataset is \$162,397.
- *Logical inconsistencies.*
 - 5 customers have spend but no recorded purchases.
 - 3 report more deal purchases than total purchases.
 - 10 of the 18 lowest incomes are exactly \$7,500, suggesting a default value.

Univariate analysis.

- The target is imbalanced, as 14.9% of training customers responded. A model that always predicts "no response" would be 85.1% accurate while finding no responders.
- *Spend is heavily skewed and often zero*
 - All six spend columns are right-skewed (skew 1.13 to 2.09), with a long tail of genuine high spenders rather than errors.
 - Nearly one in five customers bought no fruit (17.8%), fish (16.8%), or sweets (18.6%), while almost everyone bought wine and meat.
- *Some categories are very small.* Graduation is the most common education level (906 customers) and Basic the smallest (45). Married (685) and Together (475) dominate marital status, while Alone, Absurd, and YOLO have 2 customers each.

Bivariate analysis against Response. We compared response rates across groups against the overall rate of 14.9%.

- *Earlier campaign acceptance is the strongest pattern.*
 - Customers who accepted any single earlier campaign responded at 36.3% to 68.0%, versus 11.9% to 14.1% for those who didn't.
 - Response climbs steadily with the number of earlier campaigns accepted: 8.1% with none, 31.8% with one, 47.8% with two, and 79.2% with three or more.

- Still, 43.1% of responders had never accepted an earlier campaign.
- *Household composition findings:*
 - Single (23.1%), widowed (22.6%), and divorced (21.9%) customers respond at roughly twice the rate of married (10.9%) and cohabiting (9.9%) customers.
 - Customers with no teenager at home respond at 20.0%, versus 9.4% with one.
 - Response falls from 16.8% with no young children to 4.8% with two.
- For education level, response rises from 4.4% (Basic) to 20.1% (PhD).
- *Complaints show little difference* (17.6% vs. 14.9%), based on only 17 customers.

Multivariate analysis. We used Spearman correlation on the numeric features plus nine derived features: Age, Tenure_days, Children, TotalSpend, TotalPurchases, PriorAccepts, and three ratios (WineShare, DealShare, SpendPerPurchase).

- *Income, spend, and purchases largely measure the same thing: affluence and engagement.* Examples include Income and wine spend (0.84), meat spend and catalog purchases (0.85), and wine and meat spend (0.82). Among the derived features, TotalSpend and SpendPerPurchase are almost identical (0.98).
- *Young children and frequent website visits are both associated with lower income and spend* (about -0.6).
- Recency shows essentially no correlation with any other feature, and Tenure_days only weak correlation (0.3 or below).

3.2 Insights: what the EDA tells us about the problem

Response rates range from 8.1% among customers who never accepted a campaign to 79.2% among those who accepted three or more. Responders are therefore identifiable in advance, so contacting fewer, better-chosen customers is realistic.

However, because 43.1% of responders had never accepted an earlier campaign, a rule based only on past acceptance would miss a large share of them. The model will need to draw on other signals to find these customers, such as marital status, children at home, and education.

With 85.1% non-responders, a model that finds no one still looks accurate. We will evaluate on recall, precision, and PR-AUC, and report how many responders are reached when contacting the top-ranked share of customers, which is the business question itself.

Key challenges.

- *Few positives.* Only 267 responders in training, so performance estimates across cross-validation folds will be noisy.
- *Label noise.* 16 duplicate groups in training carry conflicting labels that no model can resolve.
- *Redundant features.* Highly correlated income, spend, and purchase columns make linear-model coefficients unstable and feature importance harder to interpret.
- *Small groups.* Rates for groups such as Basic education (45 customers) or two young children at home (42) are unreliable on their own.

3.3 Data problems and pre-processing recommendations

Remove uninformative columns. Drop ID, Z_CostContact, and Z_Revenue as features.

Handle duplicates: Keep group-aware splitting (StratifiedGroupKFold) in cross-validation so copies of the same customer never appear on both sides. Test whether removing the conflicting profiles changes results.

Fix invalid values, then impute.

- Treat the \$666,666 income and birth years before 1920 as missing.
- Impute Income with the training median, and add a missing-value indicator.
- Flag the ten \$7,500 incomes and test whether treating them as missing helps.

Clean up categories. Merge Alone into Single, and Absurd and YOLO into Single or an "Other" group. Consider merging small education levels, then one-hot encode.

Convert and engineer features.

- Parse Dt_Customer into Tenure_days and replace Year_Birth with Age.
- Add PriorAccepts and Children, given the clear response differences by earlier acceptance and household composition.

Manage correlated features. Use regularised logistic regression or tree-based models, and avoid including both TotalSpend and SpendPerPurchase.

Account for class imbalance. Use stratified cross-validation, class_weight="balanced" or threshold tuning, and imbalance-aware metrics rather than accuracy.